

Bridgeless toroidal maps

A square convolution for OEIS A343093

Section 2 of *Three advances on OEIS conjectures and asymptotics*

Prepared with ChatGPT

Building on Vladimir Reshetnikov's ProveIt research repository

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Abstract

We prove the square-convolution generating function posted as a conjecture for the bridgeless toroidal maps of OEIS A343093, and derive positive coefficient formulas and a three-term asymptotic expansion. *[write]* The delivered abstract continues with the manuscript's two other subjects, the nonlinear reversion pair A088714 and A088713 and the unitary-divisor partitions A301981 and A301982. Those parts are printed in two other ProveIt reports (Section 3); this report prints the manuscript's introduction, its whole map section, and the parts of its verification and research sections that concern the maps.

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1 Results, provenance, and the proof boundary

This article develops three advances on concrete OEIS questions. The first proves a generating-function identity explicitly marked conjectural in A343093. The second answers three analytic and asymptotic questions about A088714 and A088713 in the ProveIt research corpus. The third strengthens an existing refutation of two partition asymptotics to quantitative oscillation in both directions. The subjects differ, but the proof strategy has a common feature: an exact structural identity is retained long enough to control information that a leading estimate loses.

The repository snapshot used throughout is

eaf08931cbfd0132dd36ab82a7743ccb02506dc7, 4 October 2026, 00:59:12 UTC.

The related OEIS entries and the cited primary literature were inspected on 4 October 2026 UTC. The manuscript supplies written proofs, independent internal mathematical reviews, exact finite checks, and an interval certificate. These are distinct forms of evidence: a finite computation does not establish an infinite asymptotic statement. External peer review and proof-assistant formalization remain future work.

1.1 A map identity, with all root multiplicities included

If b_n counts rooted toroidal maps with n edges and no bridges, then

$$\sum_{n \geq 2} b_n z^n = \left(\frac{1}{4} \sum_{n \geq 1} \binom{4n}{n} z^n \right)^2. \quad (1.1)$$

This is Peter Bala’s posted conjecture [1]. The new derivation applies the bridge decomposition to the classical planar and toroidal generating functions of Bender–Canfield–Robinson. The genus carried by a branch changes the substituted argument, producing an essential derivative term. Keeping that term gives the exact square. The resulting coefficient formula also yields

$$b_n = \left(\frac{256}{27} \right)^n \left\{ \frac{1}{24} - \frac{7\sqrt{6}}{216\sqrt{\pi n}} + O(n^{-3/2}) \right\}.$$

The article includes the next explicit correction, an algebraic equation, and an independent enumeration of rotation systems at small sizes.

1.2 The manuscript’s other two subjects

[write] **Added 3 October 2026, batch 86.** The delivered Sections 1.2 (“Three answers for the golden-ratio moment pair”) and 1.3 (“Partition asymptotics with arbitrarily large errors of both signs”) summarize results proved in the manuscript’s Sections 3–6 and 7. Those sections are not printed in this report, so the two summaries are replaced by this note; Section 3 says where the results are printed. In brief: for the reversion pair A088714/A088713, the manuscript proves that the moment laws of the prior ProveIt report [4] have strictly positive real-analytic densities on $(0, \infty)$ with sharp equivalents at the origin, a nonzero real-analytic antiperiodic profile in the first correction to the golden-ratio transform law, and a fixed-shift renewal expansion of the companion coefficients; for A301981/A301982 it proves that each coefficient-to-model ratio has lower limit 0 and upper limit ∞ , strengthening the refutation in [5]. The delivered equations (1.2)–(1.4) went with these summaries.

1.3 What the novelty claim means

The map assertion is a proof of a currently posted conjectural identity; the underlying map formulas and bridge decomposition are credited classical ingredients. The moment-law existence theorem, leading golden-ratio equivalents, and coefficient ratio law are prior ProveIt results. The new statements resolve that report’s Questions 23.1, 23.3, and 23.6. The two-sided partition theorem answers its predecessor’s Section 11, question 3; it does not claim the earlier refutation as new. The general methods of analytic inversion, Lagrange inversion, and Landau positivity are also established methods.

The search did not identify earlier proofs of these particular new conclusions. It does not establish exhaustive historical priority, and older map-enumeration transformations may encode equivalent identities. Each Part states its dependencies explicitly so that its correctness can be assessed independently of any claim about importance or novelty.

[write] **Added 3 October 2026, batch 86: provenance, notation and the repository.** This report prints part of one manuscript: manuscript 01 of batch 86 of ProveIt’s incoming-reports intake, *Three advances on OEIS conjectures and asymptotics*, delivered as the archive `oeis_advances.zip` (580,552 bytes, wrapper directory `oeis_advances/`, 11 files) in commit `ae9baa422` and placed in commit `0f084afa9` (batch 86A) in the research-report collection, among the enumerative-combinatorics reports, as `a343093-bridgeless-toroidal-maps`. The title page reads “Prepared with ChatGPT”. The manuscript treats three subjects and was split by subject at placement: its Section 2 (its Part I) is this report; its other sections went to two existing reports (Section 3). Printed here unchanged are the introduction’s opening, Section 1.1, Section 1.4 (which speaks for the whole manuscript and is kept entire because it carries the novelty boundary), and the whole of Section 2 (every theorem, equation and subsection number of Section 2 is the delivered one). The manuscript’s Sections 8 and 9 are printed here as Sections 4 and 5, reduced to the material on maps plus the statements that concern the whole manuscript. Every label carries the prefix `btm::`; the title, abstract and running head were adapted; dated notes such as this one were added. “Part I” in the delivered text means Section 2 here; “Parts II and III” are the material printed elsewhere.

Pin and repository claims. The snapshot quoted above, `eaf08931c`, is the ProveIt commit the manuscript was written against. At placement and again at this writing, a search of the tracked tree for A343093, “bridgeless”, “rooted map”, “chord diagram”, Bender–Canfield–Robinson and the binomial $\binom{4n}{n}$ found nothing outside this report; the statement at the end of Section 2 that the inspected snapshot has no A343093 match remains true. The report continues no repository material. No statement of this report is formalized in Lean or Rocq, and its place in the collection confers no formal status. A general formal Lagrange–Bürmann theorem exists in ProveIt, `Fabius.Lagrange.coeff_solution_subst_derivative` in `Analysis/FabiusFunction/Lean/FabiusFunction/LagrangeInversion.lean`; the two Lagrange-inversion steps of Section 2 have the form it states (weight $(1+w)^4$), but nobody has instantiated it for them, and nothing formal concerns maps.

Notation. The letters of Section 2 are local to it. b_n is A343093 (rooted toroidal bridgeless maps with n edges); $m_{g,n}$, M_g , $M(z,u)$ count all rooted maps of genus g ; $B_g(t)$, $B(t,u)$ count nonempty bridgeless ones, so $B_1(t) = \sum b_n t^n$. The variable z marks edges of all maps and t edges of bridgeless cores (in (1.1) the delivered introduction writes z where (2.1) writes t ; the series are the same), while u marks genus and is not the auxiliary parameter U of (2.5) and (2.10). The letter R is defined twice: $R = \sqrt{1-12z}$ in (2.2), and $R = (1-3U)/(1+3U)$ in the proof of Theorem 2.4, which says it “reuses” the letter. Under $z = U/(1+3U)^2$ the two expressions coincide, but in that proof R is a function of t through $U(t)$, not of z . P in (2.4) is $1+B$, Q and Q_0 are corner factors, and $\rho = 27/256$ is the radius of convergence of B_1 ; in the reports of Section 3 the same letters have other meanings. No symbol was renamed.

2 A square convolution for bridgeless toroidal maps

Let b_n be the number of rooted toroidal maps with n edges and no isthmuses. These are the numbers in OEIS A343093, with $b_0 = b_1 = 0$. The entry attributes the following generating-function conjecture to Peter Bala, dated July 24, 2025 [1]:

$$\sum_{n \geq 2} b_n t^n = \left(\frac{1}{4} \sum_{n \geq 1} \binom{4n}{n} t^n \right)^2. \quad (2.1)$$

We prove this identity from the classical generating functions for all planar and toroidal maps. The proof also gives a positive finite formula, an algebraic equation, and an expansion beyond the leading asymptotic term. The result is a resolution of the explicitly stated OEIS conjecture; the underlying enumeration of unrestricted toroidal maps and the decomposition along bridges are established ingredients.

2.1 Conventions and the classical inputs

A map is a finite connected graph, with loops and parallel edges permitted, cellularly embedded in a closed oriented surface. Cellular means that every component of the complement is an open disk. Isomorphisms preserve the orientation. A root is a distinguished directed edge, or dart. The one-vertex edgeless planar map is included by convention and has one root corner. An isthmus, also called a bridge, is an edge whose deletion disconnects the underlying graph. A loop is consequently never a bridge. In particular, the restriction here is on edge connectivity: cut vertices and separating cycles are allowed.

Write $m_{g,n}$ for the number of rooted maps of genus g with n edges, and put

$$M_g(z) = \sum_{n \geq 0} m_{g,n} z^n, \quad M(z, u) = \sum_{g \geq 0} u^g M_g(z).$$

For a fixed n , only finitely many genera occur, so that these operations can be performed in formal power series without an analytic convergence assumption. Let $B_g(t)$ count the nonempty bridgeless rooted maps of genus g , and let $B(t, u) = \sum_{g \geq 0} u^g B_g(t)$. In particular, $B_1(t) = \sum_{n \geq 2} b_n t^n$.

The two external enumerative inputs are

$$M_0(z) = \frac{4(2R+1)}{3(R+1)^2}, \quad M_1(z) = \frac{(R-1)^2}{12R^2(R+2)}, \quad R = \sqrt{1-12z}, \quad R(0) = 1. \quad (2.2)$$

They are equations (4.20) and (4.22), respectively, on printed page 268 of Bender, Canfield, and Robinson [2]. Their convention of distinguishing an edge, its direction, and its side up to arbitrary surface homeomorphism is equivalent, for orientable surfaces, to choosing an orientation and a dart up to orientation-preserving homeomorphism. Thus no factor of two is needed in (2.2).

2.2 Removing bridges, with the root accounted for

Proposition 2.1 (Bridge decomposition). *With the preceding conventions,*

$$M(z, u) = 1 + zM(z, u)^2 + B\left(\frac{z}{(1 - zM(z, u))^2}, u\right). \quad (2.3)$$

Proof. The edgeless map contributes 1. If the root edge is a bridge, cut its edge ribbon in a regular neighborhood of the embedded graph and cap the boundary circles of the resulting two ribbon submaps. The components are rooted at the corners determined by the former root edge, and their order is determined by its direction. A rooted corner in a nonempty oriented map is equivalent to a rooted dart. Either component may be the edgeless planar map. Conversely, any ordered pair of rooted maps can be joined at their root corners by a new directed bridge. This is an inverse construction and contributes zM^2 .

Suppose instead that the root is not a bridge. Remove all bridges and retain the connected component containing the root. It is a nonempty bridgeless rooted core, say with k edges, and it has exactly $2k$ corners. Every component discarded from the core is attached to it through a bridge, possibly with further bridges farther from the core. At each corner, the cyclic order of darts at the incident vertex gives a linearly ordered sequence of branches. A branch consists of its attaching bridge and the rooted map on its far side. Its generating function is zM , so one corner contributes

$$1 + zM + (zM)^2 + \cdots = \frac{1}{1 - zM}.$$

Every one of the $2k$ corners, including the corner adjoining the root, admits this construction. The root remains a dart of the core; there is no further choice of a corner or a new root. A core of size k therefore has weight $z^k(1 - zM)^{-2k}$.

The decomposition and reconstruction are unique. Indeed, graph bridges are intrinsic, the component containing the nonbridge root is intrinsic, and the rotations at core vertices recover both the relevant corners and the order of branches. Conversely, grafting ordered branches into these corners creates precisely the bridges that are removed by the procedure.

It remains to check the genus weight. Joining two cellular maps of genera g_1, g_2 by a bridge produces a map with

$$V = V_1 + V_2, \quad E = E_1 + E_2 + 1, \quad F = F_1 + F_2 - 1.$$

Euler's formula gives $2 - 2g = 2 - 2(g_1 + g_2)$. Thus genera add under every bridge attachment and also in the root-bridge case. Multiplying the weights u^{g_i} is correct. Summing the core contribution proves (2.3). $\square$

This is the usual canonical decomposition into a bridgeless root component and branches. Courtiel, Yeats, and Zeilberger give a closely related formulation in Proposition 23 [3]. Their auxiliary dangling root is included in the size, whereas our root is one of the ordinary edges. The proof above fixes the normalization and explains the exponent $2k$ in (2.3) directly.

There is also a useful inverse form. If $P = 1 + B(t, u)$, then

$$M = P(1 + tP), \quad z = \frac{t}{(1 + tP)^2}. \quad (2.4)$$

To see this, set $Q = 1 - zM$ in (2.3). It gives $P = MQ$ and $t = z/Q^2$, whence $Q = 1/(1 + tP)$. These identities are formal: the relevant series have constant term 1, and $t = z + O(z^2)$ is invertible for composition.

2.3 Extracting genus one

Introduce the auxiliary variable

$$U = \frac{1 - R}{3(1 + R)}, \quad z = \frac{U}{(1 + 3U)^2}. \quad (2.5)$$

The planar and toroidal formulas (2.2) become

$$M_0 = (1 - U)(1 + 3U), \quad M_1 = \frac{U^2(1 + 3U)}{(1 + U)(1 - 3U)^2}. \quad (2.6)$$

For the moment U is viewed as a formal series in z . At genus zero set

$$Q_0 = 1 - zM_0 = \frac{(1 + U)^2}{1 + 3U}, \quad t_0 = \frac{z}{Q_0^2} = \frac{U}{(1 + U)^4}.$$

The genus-zero coefficient of (2.3) then gives

$$B_0(t_0) = M_0 - 1 - zM_0^2 = U - U^2 - U^3. \quad (2.7)$$

Since

$$\frac{d}{dU}(U - U^2 - U^3) = (1 + U)(1 - 3U), \quad \frac{dt_0}{dU} = \frac{1 - 3U}{(1 + U)^5},$$

we obtain the particularly simple identity

$$B'_0(t_0) = (1 + U)^6. \quad (2.8)$$

Now expand (2.3) to first order in u . The coefficient of u in its substituted argument is

$$[u] \frac{z}{(1 - zM(z, u))^2} = \frac{2z^2 M_1}{Q_0^3}.$$

Consequently

$$B_1(t_0) = M_1 \left(1 - 2zM_0 - \frac{2z^2 B'_0(t_0)}{Q_0^3} \right). \quad (2.9)$$

There are no other genus-one terms: either the core has genus one and all branches are planar, or the core is planar and exactly one branch has genus one. Substituting the explicit formulas into the parenthesis gives

$$1 - \frac{2U(1 - U)}{1 + 3U} - \frac{2U^2}{1 + 3U} = \frac{1 + U}{1 + 3U}.$$

Together with (2.6), this proves the following theorem.

Theorem 2.2 (Exact toroidal generating function). *Let $U(t)$ be the unique formal series with $U(0) = 0$ satisfying*

$$U = t(1 + U)^4. \quad (2.10)$$

Then

$$B_1(t) = \frac{U(t)^2}{(1 - 3U(t))^2}. \quad (2.11)$$

In particular, Bala's identity (2.1) holds for every coefficient.

Proof. Equation (2.9) proves (2.11), because $t_0 = z + O(z^2)$ may be used as a formal variable. For the claimed square, differentiate (2.10) to obtain

$$\frac{U}{1 - 3U} = t \frac{d}{dt} \log(1 + U).$$

Lagrange inversion gives, for $n \geq 1$,

$$[t^n] \log(1 + U(t)) = \frac{1}{n} [u^{n-1}] (1 + u)^{4n-1} = \frac{1}{4n} \binom{4n}{n}.$$

It follows that $U/(1 - 3U) = \frac{1}{4} \sum_{n \geq 1} \binom{4n}{n} t^n$. Squaring establishes (2.1). $\square$

2.4 Exact coefficients and an algebraic certificate

Corollary 2.3. *For $n \geq 2$,*

$$b_n = \frac{1}{16} \sum_{k=1}^{n-1} \binom{4k}{k} \binom{4(n-k)}{n-k}, \quad (2.12)$$

$$b_n = \frac{1}{n} \sum_{j=0}^{n-2} (j+1)(j+2)3^j \binom{4n}{n-2-j}. \quad (2.13)$$

Proof. The first formula is the Cauchy product in (2.1). For the second, apply Lagrange inversion to $U^2/(1-3U)^2$; its derivative with respect to U is $2U/(1-3U)^3$. Thus

$$b_n = \frac{2}{n} [u^{n-2}] \frac{(1+u)^{4n}}{(1-3u)^3}.$$

Expanding the last denominator gives (2.13). □

The first ten values, starting at $n = 2$, are

$$1, 14, 159, 1680, 17147, 171612, 1696491, 16631840, 162090756, 1572801142.$$

The first factor in the convolution is integral: $\frac{1}{4} \binom{4k}{k} = \binom{4k-1}{k-1}$. Hence (2.12) also gives an elementary positive model for the numbers: ordered pairs of binary words of respective lengths $4k-1$ and $4(n-k)-1$, containing $k-1$ and $n-k-1$ ones. The enumeration proves this equinumerosity; constructing a direct geometric bijection is a further question.

For an explicit algebraic certificate put $A = B_1(t)$ and $H = U/(1-3U)$. Solving for U yields

$$t(1+4H)^4 = H(1+3H)^3, \quad A = H^2.$$

Separating even and odd powers of H gives

$$\left[t + (96t-9)A + (256t-27)A^2 \right]^2 - A[(16t-1) + (256t-27)A]^2 = 0. \quad (2.14)$$

The intended solution is the one specified by (2.10) and (2.11). Equation (2.14) alone is an elimination relation and should not be used to choose a branch without these initial conditions.

2.5 Growth and its first corrections

Theorem 2.4 (Asymptotic expansion). *With $\rho = 27/256$, as $n \rightarrow \infty$,*

$$b_n = \rho^{-n} \left[\frac{1}{24} - \frac{7\sqrt{6}}{216\sqrt{\pi}} n^{-1/2} + \frac{217\sqrt{6}}{93312\sqrt{\pi}} n^{-3/2} + O(n^{-5/2}) \right]. \quad (2.15)$$

A full expansion in the successive powers $n^{-1/2}, n^{-3/2}, \dots$ follows by the coefficient procedure in the proof. In particular, $b_n \sim (256/27)^n/24$.

Proof. The characteristic equation for $U = t(1+U)^4$ is $1 = 4t(1+U)^3$. It has the positive solution $U = 1/3$, $t = \rho$. Equivalently, $t = U/(1+U)^4$ has its only finite critical value away from $t = 0$ at ρ . The branch $U(0) = 0$ is analytic at zero and admits analytic continuation to a disk slightly larger than $|t| < \rho$, slit along the positive real ray starting at ρ . One can see this directly from the algebraic equation: multiple finite roots require $U = 1/3$ and $t = \rho$, and the leading coefficient can

vanish only at $t = 0$, where this particular branch is already regular. Thus the singularity at ρ is the unique dominant one for this branch.

For explicit calculations it is convenient to reuse $R = (1 - 3U)/(1 + 3U)$. Then

$$\frac{t}{\rho} = \frac{16(1-R)(1+R)^3}{(R+2)^4}, \quad B_1(t) = \frac{(1-R)^2}{36R^2}.$$

Set $w = \sqrt{1 - t/\rho}$, positive for $0 < t < \rho$. The first relation has a convergent local inverse of the form

$$R = \frac{\sqrt{6}}{3}w + \frac{1}{9}w^2 + \frac{19\sqrt{6}}{648}w^3 - \frac{11}{972}w^4 + \frac{677\sqrt{6}}{93312}w^5 + O(w^6).$$

The existence of this inverse follows by writing $1 - t/\rho = R^2(3/2 + O(R))$ and applying the analytic inverse-function theorem to $R\sqrt{3/2 + O(R)}$. Substitution gives

$$B_1(t) = \frac{1}{24w^2} - \frac{7\sqrt{6}}{216w} + \frac{83}{2592} + \frac{161\sqrt{6}}{46656}w - \frac{1249}{279936}w^2 + O(w^3). \quad (2.16)$$

For nonintegral α the exact coefficient identity

$$[t^n](1 - t/\rho)^\alpha = \rho^{-n} \frac{\Gamma(n - \alpha)}{\Gamma(-\alpha)\Gamma(n + 1)}$$

and the gamma-ratio expansion convert the nonanalytic terms in (2.16) into coefficient asymptotics. Local analytic terms contribute no algebraic-order term from the dominant singularity. In particular,

$$\begin{aligned} [t^n](1 - t/\rho)^{-1/2} &= \frac{\rho^{-n}}{\sqrt{\pi n}} \left(1 - \frac{1}{8n} + O(n^{-2})\right), \\ [t^n](1 - t/\rho)^{1/2} &= -\frac{\rho^{-n}}{2\sqrt{\pi} n^{3/2}} \left(1 + O(n^{-1})\right). \end{aligned}$$

The w^{-2} term has coefficient exactly $\rho^{-n}/24$. The coefficient of $\rho^{-n}n^{-3/2}$ is

$$\frac{7\sqrt{6}}{1728\sqrt{\pi}} - \frac{161\sqrt{6}}{93312\sqrt{\pi}} = \frac{217\sqrt{6}}{93312\sqrt{\pi}}.$$

This proves (2.15). Continuing the convergent Puiseux inversion, followed by the gamma-ratio expansion for each odd power of w , gives any prescribed number of terms, with a remainder of the corresponding order. The slit-domain continuation supplies the usual coefficient-transfer justification. $\square$

For comparison, all toroidal maps satisfy $m_{1,n} \sim 12^n/24$, as follows from (2.2). Hence the probability that a uniformly selected rooted toroidal map with n edges has no bridges satisfies

$$\frac{b_n}{m_{1,n}} \sim \left(\frac{64}{81}\right)^n. \quad (2.17)$$

This describes the exponential effect of the connectivity restriction.

2.6 Verification and further questions

The accompanying program `code/verify_maps.py` checks the convolution and Lagrange formulas against all twenty terms printed in the inspected OEIS entry, and against each other for additional indices. It also contains an independent enumeration for small maps. Fix the edge involution on $2n$ darts to be $(0\ 1)(2\ 3)\cdots(2n-2\ 2n-1)$ and enumerate every vertex permutation σ . Connectivity is tested in the underlying multigraph; the cycles of σ and $\sigma\alpha$ give vertices and faces. Euler’s formula determines the genus, and deleting each edge tests the bridge restriction. Keeping dart 0 as root, each isomorphism class is counted exactly $2^{n-1}(n-1)!$ times: these are precisely the relabelings commuting with the fixed involution and fixing the root. Their action on connected rooted maps is free. Thus division by this known factor gives a genuinely separate check of the normalization and the first coefficients. These finite checks support the implementation; the all-index identity is established by Theorem 2.2.

Several questions now have concrete starting points.

1. Find a direct bijection underlying the square in (2.1). The product describes two independently chosen objects of positive sizes, and a geometric interpretation should explain how a toroidal bridgeless map yields that ordered pair.
2. Refine the square identity by vertices and faces, using the two-variable map enumeration and OEIS A343092. A valid refinement must reduce to (2.1) after summing the face parameter; an arbitrary weighting of the two factors need not count maps.
3. Carry (2.3) to genus two and higher. At each genus the new B_g enters linearly after all smaller genera are known, so the transformation gives an effective route from unrestricted-map formulas to bridgeless-map formulas. Determining which genera admit comparably simple positive factorizations is a separate question.
4. Mark bridges with a variable q . The same decomposition gives $M = 1 + qzM^2 + B(z/(1 - qzM)^2, u)$, with M now recording the bridge count. This supplies exact generating functions from which to study the distribution of the number of bridges and of the core sizes.

The bounded source audit found the conjecture still explicitly marked as such in the inspected OEIS entry and found no A343093 match in the inspected ProveIt snapshot. It also identified the classical sources for both crucial inputs. This supports the precise description “a proof of the posted OEIS identity”; it does not establish that every equivalent specialization is absent from the older map-enumeration literature.

3 The manuscript’s Sections 3–7: where they are printed

[write] **Added 3 October 2026, batch 86.** The manuscript’s Parts II and III, Sections 3–7, continue two existing ProveIt reports and were placed with them in commit 0f084afa9. They are printed there, not here; the Parts named below were being written at the same time as this report, so their labels are not cited. Paths are relative to `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/`.

- *Sections 3–6* (Part II, “The nonlinear reversion pair A088714 and A088713”: the inherited foundation, the density theorem for the moment laws μ and ν , the complete golden-ratio expansion with its interval certificate, and the fixed-shift renewal expansion of the companion coefficients) answer Research questions 23.1, 23.3 and 23.6 of `a088714-bell-scale-growth` [4] and are printed there as its planned **Part IV**, with the programs `code/04-densities-profile-verify_dynamics.py` and `code/04-densities-profile-verify_renewal.py` and their recorded outputs in `data/`. The rows of the manuscript’s Section 8 on these programs, its Section 8.2 (the certificate’s role in the proof), the renewal table of its Section 8.3, and its research directions 9.1–9.3 and 9.6 go with them.
- *Section 7* (Part III, “Two-sided fluctuations of unitary-divisor partitions”) answers question 3 of Section 11 of `a301981-unitary-divisor-partitions` [5] and is printed there as its planned **Part II**, with `code/02-two-sided-verify_partitions.py` and `data/02-two-sided-partition_verification.json`, together with the partition rows of Section 8 and research direction 9.5.

The bibliography entries used only by those sections (A088713, A088714, A301981, A301982, A034448, Huang–Wang, Python’s `decimal` documentation, Landau, Diamond–Pintz, Tóth, Trudgian and the DLMF) go with them; the two ProveIt reports stay in the bibliography here because Section 1 cites them.

A statement of the manuscript that was stale on arrival. The manuscript’s Section 9.1 and its delivery README say that the finer Bell-number normalization of A088714 (Conjecture 8.1 of `a088714-bell-scale-growth`) remains open. That was true at the pin, but manuscript 02 of the same batch (`oeis_research_bundle.zip`, same arrival commit, pin b7e4f25b6) gives a proof of it, placed as the planned Part V of that report. That proof is AI-assisted and unrefereed; the intake read its argument without finding a gap but asks for an independent review before the conjecture is called settled. Nothing in this report depends on it.

Shared statements kept here. The statements of the manuscript that concern all three of its subjects are kept in this report: the evidence boundary of Section 1 (written proofs, internal reviews, exact finite checks and an interval certificate are distinct; a finite computation does not establish an infinite asymptotic statement; external review and formalization are future work), the novelty boundary of Section 1.4, the review disclaimer of Section 4, and the submission statement of Section 5.2 (no OEIS update or other submission was made).

4 Verification, reproducibility, and interpretation

The accompanying archive contains the complete L^AT_EX source, the PDF, four verification programs, their exact or interval output, and a short reproduction guide. The programs do not access the network. The stored OEIS prefixes and the repository pin make their finite reference data explicit. Standard-library integer and rational arithmetic suffice for the renewal and partition checks. The dynamics certificate additionally uses correctly rounded decimal elementary functions with outward enclosure. The map program uses SymPy for exact symbolic identities and mpmath only for numerical asymptotic diagnostics.

4.1 What was checked

Program	Verification performed
<code>verify_maps.py</code>	Checks both positive coefficient formulas against all 20 displayed OEIS terms; compares the formulas through $n = 100$; independently enumerates connected rotation systems through four edges; verifies the bridge substitutions, elimination relation, and Puiseux coefficients symbolically.

[write] **Added 3 October 2026, batch 86: the shipped program.** The delivered table has four rows; the rows for `verify_dynamics.py`, `verify_renewal.py` and `verify_partitions.py` are printed with the material they check (Section 3). This report ships `code/verify_maps.py` under its delivered name and its recorded output, delivered as `verification/maps_checks.json`, as `data/maps_checks.json`. The program writes to `../verification/maps_checks.json` unless given `-output`; the report’s README gives a rerun that leaves the shipped files untouched. On 3 October 2026 the intake reran it on a scratch copy (SymPy 1.14.0, mpmath 1.3.0, Python 3.13.5, Windows): it passed in about 30 s, and its output equals the shipped file after removal of carriage returns. The intake also checked, independently, that the power series of $U^2/(1 - 3U)^2$ with $U = t(1 + U)^4$ agrees with (2.12) through t^{59} , and that $((27/256)^n b_n - (\text{three-term approximation}))n^{5/2}$ is $-2.2132 \cdot 10^{-5}$ at $n = 1000$ and $-2.2079 \cdot 10^{-5}$ at $n = 4000$, consistent with the error term of (2.15); like the delivered diagnostics, these are checks, not proofs.

Separate mathematical reviews checked the root normalization and genus derivative in Part I; the global-inversion hypotheses, boundary limits, and zero descent in the density theorem; the invariant coordinates, analytic reversion, and interval enclosure in the golden-ratio expansion; the convolution bounds in the renewal theorem; and the noncancellation, positivity argument, saddle bridge, and inverse scaling in Part III. These reviews are internal checks of a new manuscript. They are not external refereeing or machine-checked formal proofs.

4.2 Diagnostics that do not enter the proofs

[write] **Added 3 October 2026, batch 86.** The delivered Section 8.2 (“A certificate with a mathematical role”) and the first part of Section 8.3 (the renewal diagnostics and their table) concern Sections 3–6 and are printed with them; the delivered closing paragraph on the partition theorem goes with Section 7. The map diagnostics of Section 8.3 follow, with the sentence that introduces them adapted.

The maps have a visible $n^{-1/2}$ correction: $(27/256)^n b_n$ is approximately 0.03376725, 0.03771030, and 0.03968765 at $n = 32, 128, 512$, respectively, while its limit is $1/24 \approx 0.04166667$. The three-term approximation has the proved error $O(n^{-5/2})$; the numerical agreement is only a check of the implemented constants.

5 A proposed research program

The following directions are ordered by how directly they continue the proved structures. Each states a concrete objective and the obstacle that remains. The local questions at the end of Parts I and III supply additional variants.

[write] **Added 3 October 2026, batch 86.** Of the delivered seven directions, Sections 9.1–9.3 and 9.6 (the finer Bell comparison, the oscillation profile, effective tail laws, the two general mechanisms) and Section 9.5 (effective partition oscillations) are printed with the material they continue (Section 3). Section 9.4 and Section 9.7 follow; the local questions of Part I are those at the end of Section 2.6. Section 9.7 is printed entire because it speaks for all three subjects.

5.1 Find a direct map bijection and extend the genus calculation

The exact square has the integral factor $\frac{1}{4}\binom{4n}{n} = \binom{4n-1}{n-1}$. This suggests a direct decomposition of a bridgeless rooted toroidal map into a pair of independently described objects of total size n . A bijection should explain the root and the positive convolution, rather than merely reproducing the same algebraic generating function. It would also clarify whether a face or vertex statistic has a natural refinement on the two factors.

The multigenus bridge equation already contains a separate algebraic research program. Insert known unrestricted genus- g map series, extract the coefficient of the genus marker, and retain every derivative caused by the substituted argument. Determine rational forms in the common planar parameter and systematic singular exponents for the bridgeless series. Higher-genus expressions are not obtained by a naive substitution into the unrestricted series; Part I displays the missing contribution already at genus one.

5.2 Formalization and an independent submission path

The exact bridge decomposition, the positive coefficient formulas, and the renewal convolution estimates are comparatively self-contained targets for formalization. The interval certificate can be ported to a small formally checked rational-enclosure kernel. The density and Mellin arguments require a broader analytic foundation, making them later targets.

For external mathematical review, the three Parts can be separated into articles with their dependencies preserved. An OEIS update for A343093 should cite the complete rooted-map proof and its normalization; updates for the partition entries should distinguish the earlier refutation from the new two-sided strengthening. No update or submission has been made as part of this work. A bibliographic comparison with the older map-enumeration literature and independent refereeing remain appropriate before asserting historical priority.

[write] **Added 3 October 2026, batch 86.** The ProveIt intake made no OEIS submission either. The split recommended in the second paragraph is the one made at placement (Section 3). The formal Lagrange–Bürmann theorem named in the note at the end of Section 1 would be a starting point for the coefficient formulas; the bridge decomposition itself has no formal counterpart in ProveIt.

6 Provenance

[write] **Added 3 October 2026, batch 86.** *One manuscript, partly printed.* Batch 86, manuscript 01: *Three advances on OEIS conjectures and asymptotics. Bridgeless toroidal maps, golden-ratio moment laws, and unitary-divisor partitions*, “Prepared with ChatGPT”, dated 4 October 2026 (UTC); archive `oeis_advances.zip`, arrival commit `ae9baa422`; pin `eaf08931c` (3 October 2026, 17:59:12 Pacific time; the full hash is quoted in Section 1); placement commit `0f084afa9` (batch 86A). It contributed the whole of this report: Sections 1 and 2, and Sections 4 and 5 from its Sections 8 and 9.

Where the write had to choose. Only in dividing the shared sections: the introduction’s summaries of the other two subjects were replaced by a note; the verification table, diagnostics and research directions were divided by subject; every statement that concerns the whole manuscript was kept. Section 2 is printed in full and unchanged apart from its labels. The delivered PDF (40 pages) is not shipped; it survives, with the full source, in the archive of the arrival commit.

References

- [1] The OEIS Foundation Inc., *A343093: Number of rooted toroidal maps with n edges and no isthmuses*, entry by Andrew Howroyd; conjectural generating function by Peter Bala, July 24, 2025. <https://oeis.org/A343093>. Accessed October 4, 2026 (UTC).
- [2] E. A. Bender, E. R. Canfield, and R. W. Robinson, *The enumeration of maps on the torus and the projective plane*, Canadian Mathematical Bulletin **31** (1988), 257–271.
<https://doi.org/10.4153/CMB-1988-039-4>. Exact inputs: equations (4.20) and (4.22), p. 268.
- [3] J. Courtiel, K. Yeats, and N. Zeilberger, *Connected chord diagrams and bridgeless maps*, Electronic Journal of Combinatorics **26**(4) (2019), Paper P4.37, 56 pp.
<https://doi.org/10.37236/7400>; <https://arxiv.org/abs/1611.04611>. See Proposition 23, pp. 17–18, with the auxiliary dangling-root normalization explained in the present text.
- [4] ProveIt research-report collection, *Bell-Scale Growth of OEIS A088714*, merged article, Parts I–III, September 20–October 2, 2026. AI-assisted unrefereed research report. Pinned source in Vladimir Reshetnikov’s ProveIt repository. Snapshot `eaf08931cbfd0132dd36ab82a7743ccb02506dc7`. The analytic foundations and Questions 23.1, 23.3, 23.6 are in Part II; Part III identifies the remaining fine Bell-comparison estimates.
- [5] ProveIt research-report collection, *Unitary divisor sum colored partitions and their asymptotics*, October 2, 2026. AI-assisted unrefereed research report. Pinned source in Vladimir Reshetnikov’s ProveIt repository. Snapshot `eaf08931cbfd0132dd36ab82a7743ccb02506dc7`.